## Surd rules

**1**

| | | | |
|---|---|---|---|
| a | False | b | True |

**2**

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| a | $\sqrt{252}$ | b | $\sqrt{224}$ | c | $\sqrt{45}$ | d | $\sqrt{1000}$ |

**3**

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| a | $4+3\sqrt{5}$ | b | $5\sqrt{2}$ | c | $45$ | d | $735$ |
| e | $6\sqrt{x}$ | f | $\sqrt{3}$ | g | $\frac{\sqrt{23}}{4}$ | | |

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## Multiplication and division of surds

**4**

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|---|---|---|---|---|---|---|---|
| a | $\sqrt{22}$ | b | $4\sqrt{2}$ | c | $20\sqrt{7}$ | d | $15\sqrt{35}$ |
| e | $\sqrt{70}$ | f | $35\sqrt{21}$ | g | $\sqrt{35}$ | h | $80\sqrt{5}$ |
| i | $\sqrt{231}$ | j | $11\sqrt{5}$ | k | $20\sqrt{11}$ | l | $30\sqrt{55}$ |
| m | $14\sqrt{11}$ | n | $24\sqrt{15}$ | o | $320\sqrt{3}$ | p | $40\sqrt{51}$ |
| q | $340\sqrt{7}$ | r | $216\sqrt{17}$ | | | | |

**5**

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| a | $\sqrt{5}$ | b | $2\sqrt{7}$ | c | $32$ | d | $8\sqrt{11}$ |
| e | $4\sqrt{7}$ | f | $\sqrt{3}$ | g | $\sqrt{11}$ | h | $\sqrt{3}$ |
| i | $\sqrt{7}$ | j | $\sqrt{13}$ | k | $5\sqrt{7}$ | l | $10\sqrt{5}$ |
| m | $15\sqrt{2}$ | n | $2\sqrt{7}$ | o | $3$ | p | $6$ |
| q | $10$ | r | $16$ | s | $10$ | t | $\frac{5}{9}$ |
| u | $\frac{9}{2}$ | | | | | | |

**6**

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| a | $2$ | b | $\frac{1}{\sqrt{5}}$ | c | $4$ | d | $\frac{1}{\sqrt{3}}$ |
| e | $\frac{1}{\sqrt{3}}$ | f | $2$ | g | $\frac{2}{3}$ | h | $\frac{3}{2}$ |
| i | $17\sqrt{7}$ | | | | | | |

## Mixed operations with surds

**7**

a $26\sqrt{3}$ b $-18\sqrt{5}$ c $27\sqrt{6}+11\sqrt{13}$ d $20\sqrt{6}+5\sqrt{11}$

e $4\sqrt{6}$ f $16\sqrt{5}$ g $-3\sqrt{2}$ h $42\sqrt{5}$

**8**

a $3\sqrt{3}$ b $3\sqrt{3}$ c $5\sqrt{2}$ d $6\sqrt{5}$

**9**

a $-5\sqrt{7}+\sqrt{35}$ b $7\sqrt{14}+42\sqrt{2}$

c $-42\sqrt{22}+7\sqrt{77}$ d $\sqrt{26}-3\sqrt{13}+5\sqrt{2}-15$

e $\sqrt{14}-\sqrt{77}-\sqrt{26}+\sqrt{143}$ f $12+2\sqrt{11}$

g $29+12\sqrt{5}$ h $\sqrt{77}+4\sqrt{11}$

i $3\sqrt{7}+\sqrt{21}$ j $-6\sqrt{2}+\sqrt{22}$

k $3\sqrt{39}-15\sqrt{3}$ l $\sqrt{33}+\sqrt{39}$

m $-4\sqrt{77}+4\sqrt{14}$ n $15\sqrt{11}+3\sqrt{55}$

o $-24\sqrt{14}+8\sqrt{6}$ p $15\sqrt{10}+20\sqrt{14}$

q $63\sqrt{5}$ r $11\sqrt{15}$

s $24\sqrt{77}-32\sqrt{55}$

**10**

a $60$ b $\frac{12\sqrt{3}}{\sqrt{5}}$ c $\sqrt{5}+\sqrt{7}$ d $\frac{3\sqrt{3}}{\sqrt{10}}$

**11**

a $x=5999, y=-560$ b $x=5, y=-3$

**12**

a $42+(8+2m)\sqrt{13}+m^2$ b $m=-4$

## Rationalise the denominator

**13**

a No b Yes c No

**14**

a $\frac{\sqrt{11}}{11}$ b $\frac{\sqrt{22}-2\sqrt{11}}{11}$ c $\frac{7\sqrt{12}}{12}$ d $\frac{14\sqrt{143}}{117}$

**15**

a $\frac{\sqrt{3}}{3}$ b $7\sqrt{3}$ c $\frac{-9\left(\sqrt{5}+7\right)}{44}$ d $\frac{4\left(9\sqrt{11}+5\right)}{433}$

e $\frac{9+2\sqrt{14}}{5}$ f $\frac{\sqrt{10}-1}{9}$ g $4\left(\sqrt{10}+\sqrt{3}\right)$ h $\frac{5\sqrt{42}-4\sqrt{14}}{118}$

**16**

a $-\frac{2\sqrt{2}}{23}$ b $\frac{3\sqrt{5}-15\sqrt{2}}{20}$

**17** No solution provided

**18**

a $\frac{\sqrt{3}-1}{2}$ b $16$

**19**

a $19+8\sqrt{3}$ b $\frac{6-\sqrt{3}}{3}$

**20** $\sqrt{1800}$

## Applications

**21** $12\sqrt{2}$ units

**22** $3\sqrt{5}$ units

**23**

a $4\sqrt{7}\ \mathrm{m}$ b $3\ \mathrm{m}^2$

**24** $c=\sqrt{26}$ units

**25** $9\sqrt{3}\ \mathrm{cm}$

**26**

a $\frac{\sqrt{5}+1}{2}$ b $1.618$

**27** $4\sqrt{10} - 12 \text{ m}$